MILITARY SPECIFICATION

FUZES, ROCKET, M423 AND M427
LESS BOOSTER AND BOOSTER LEAD

This Amendment forms a part of Military Specification MIL-F-50548A (AR), dated 9 November 1978, and is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

3.3.1: Delete in its entirety, and substitute the following:

"3.3.1 Arming Time - The rotor assembly in the loaded S&A shall rotate from the fully safe position to the fully armed position in not less than 0.67 sec. and not more than 0.88 sec. when subjected to arming acceleration of 27.00 plus or minus 0.25 g's. The S&A shall meet this requirement at any temperature between -65°F and +165°F."

3.4.1: Delete in its entirety and substitute the following:

"3.4.1 Arming Time - The fuze assembly shall arm in not less than 0.63 sec. or more than 0.92 sec. when subjected to arming acceleration of 27.00 plus or minus 0.25 g's and the detent shall lock the rotor in the fully armed position, as determined by radiographic inspection. The fuze assembly shall meet this requirement at any temperature between -65°F and +165°F."

3.5.1: Delete in its entirety and substitute the following:

"3.5.1 Arming Time - The rotor assembly in the loaded S&A shall rotate from the fully safe position to the fully armed position in not less than 1.13 sec. and not more than 1.37 sec. when subjected to arming acceleration of 40.00 plus or minus 0.25 g's. The S&A shall meet this requirement at any temperature between -65°F and +165°F."

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3.6.1: Delete in its entirety and substitute the following:

"3.6.1 Arming Time - The fuze assembly shall arm in not less than 1.07 sec. and not more than 1.46 sec. when subjected to arming acceleration of 40.00 plus or minus 0.25 g's and the detent shall lock the rotor in the fully armed position as determined by radiographic inspection. the fuze assembly shall meet this requirement at any temperature between -65°F and +165°F.

3.9: Delete paragraph number and substitute "3.9.1". Add new paragraph 3.9:

"3.9 Drop tests"

Add new paragraph 3.9.2:

"3.9.2 Five-foot drop. The fuze shall be safe to handle, fire, or dispose of following the five-foot drop test as specified in MIL-STD-331. It need not be operable."

3.10: Delete after "tests": "Five Foot Drop Test as specified in MIL-STD-331;".

PAGE 5

3.10, line 5: Delete "the Five Foot Drop Test and".

4.1: Delete in its entirety and substitute:

"4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification."

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PAGE 6

4 3.1: Delete "253" and subsitute "376".

PAGE 7

4.3.1: Add "or 9260776" next to "8887706". Delete "8883693" and substitute "9260804". Add "or 9260774 next to "8883692".

Add "Stud, Spring 8883721 10".

PAGE 11

TABLE 1: Delete "253" and substitute "376"; under Fuze, Rocket, Less Booster and Booster Lead, add;

EXAMINATION OR TEST	NUMBER OF SAMPLE UNITS	REQUIREMENTS PARAGRAPH	PARAGRAPH REFERENCE
Low temperature MIL-STD-810 Procedure 1, Method 502	73	3.4.1	4.5.35
High temperature MIL-STD-810 Procedure 1 Method 501	73	3.4.1	4.5.36

PAGE 12

TABLE 1: Delete "253" and substitute "376"; under Fuze, Rocket, Less Booster and Booster Lead, add;

EXAMINATION OR TEST	NUMBER OF SAMPLE UNITS	REQUIREMENTS PARAGRAPH	PARAGRAPH REFERENCE
Low temperature MIL-STD-810 Procedure 1, Method 502	73	3 6.1	4.5.35
High temperature MIL-STD-810 Procedure 1 Method 501	73	3.6.1	4.5.36

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PAGE 15

TABLE 1: Delete "(Dwg. 8883706)" and substitute "(Dwg 8883706 or 9260776)".

PAGE 16

TABLE 1: Delete "8883693" and substitute "9260804".

Delete "(Dwg. 8883692)" and substitute "(Dwg 8883692 or 9260774)".

Add the following:

"Stud, Spring (Dwg 8883721) Examination for defects	10	3.2	4.4.2.37"	

PAGE 37

4.4.2.16: Delete Note a in its entirety and substitute:

"a. Samples taken from 5 foot drop, aircraft vibration, non-environmental (including low temperature and high temperature): Note 5' drop not done on every lot."

Add:

CATEGORY	EXAMINATION OR TEST	NUMBER OF SAMPLE UNITS	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE
CRITICAL				
10	Low Temperature MIL-STD-810, Procedure 1, Method 502	73	3.4.1	4.5.35
11	High Temperature MIL-STD-810 Procedure 1, Method 501	73	3.4.1	4.5.36

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4.4.2.17: Delete Note a in its entirety and substitute: "a. Samples taken from 5 foot drop, Aircraft Vibration, Non-Environmental (including Low Temperature and High Temperature): Note 5' drop not done on every lot."

Add:

CATEGORY	EXAMINATION OR TEST	NUMBER OF SAMPLE UNITS	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE
CRITICAL				
10	Low Temperature MIL-STD-810, Procedure 1, Method 502	73	3.6.1	4.5.35
11	High Temperature MIL-STD-810 Procedure 1, Method 501	73	3.6.1	4.5.36

PAGE 40

* 4.4.2.19, Add "Major 109 Corrosion resistance of anodic coating a 3.2 4.5.10".

Add: "Note a. MIL-STD-105 Level 5.2 based on no more than 40 hours production. Parts shall be rotated 180° every 24 hours. Should one or more samples fail to meet the requirements, the lot shall be rejected."

PAGE 41

* 4.4.2.20, Major 114: Delete "visual" and substitute "Visual, 4X magnification, minimum".

* Add "Major 115 Corrosion resistance of anodic coating a 3.2 4.5.10".

Add "Note a. MIL-STD-105 Level 5.2 based on no more than 40 hours production. Parts shall be rotated 180° every 24 hours. Should one or more samples fail to meet the requirements, the lot shall be rejected."

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PAGE 44

* 4.4.2.22: Delete "Major 101" and "Note (a)" in their entirety.

PAGE 49

4.4.2.27: Under "CATEGORY" delete "1" and "2" and substitute "2" and "1", respectively.

PAGE 50

4.4.2.28: Under "CATEGORY" delete "1" and "2" and substitute "2" and "1", respectively.

PAGE 51

4.4.2.29: Under "CATEGORY" delete "1" and "2" and substitute "2" and "1", respectively.

PAGE 52

4.4.2.30: Delete in its entirety and substitute included Page 7 (Page 52 of 81).

PAGE 53

4.4.2.31: Delete in its entirety and substitute included Page 8 (Page 53 of 81).

PAGE 54

4.4.2.32: Delete in its entirety and substitute included Page 9 (Page 54 of 81).

PAGE 56

4.4.2.34: Add "or 9260774" next to "8883692".

PAGE 62

4.5.5: Add "One defective shall be cause for lot rejection".

4.5.6: Add "One defective shall be cause for lot rejection".

4.5.7.1: Delete "Apply an" and substitute "Apply the required".

Add: "and removed from the lot".

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4.5.7.2: Delete "Apply an" and substitute "Apply the required".

Add: "and removed from the lot".

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4.5.7.3: Delete "Apply an" and substitute "Apply the required".

Add "and removed from the lot".

4.5.7.4: Delete "Apply the" and substitute "Apply the required".

Add: "and removed from the lot".

4.5.7.5: Delete "Apply an" and substitute "Apply the required".

Add: "and removed from the lot".

4.5.7.6, line 5: Delete "to the outside face of the annular gear" and substitute "to the outside face of the annular gear studs".

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.30	Lever Assembly				8883706 or 9260776
					NEXT HIGHER ASSEMBLY 8883692 or 9260774
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Maximum gap between lever and shaft		100%	3.2	Gage
2	Load test of lever shaft		100%	3.2	4.5.7.4
3	Load test of lever pallet		100%	3.2	4.5.7.5
Major					
101	Torque test of lever shaft		100%	3.2	4.5.7.8
Minor					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

MIL-F-50548A (AR)
AMENDMENT 3

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8883744
4.4.2.31	Rotor Assembly				NEXT HIGHER ASSEMBLY 9215619
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Load test of annular gear		100%	3.2	4.5.7.6
Major 101	Load test of lock roller pin		0.40%	3.2	4.5.7.7
Minor 201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

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AMENDMENT 3

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9260804
4.4.2.32	Inner Plate Assembly				NEXT HIGHER ASSEMBLY 8883692 or 9260774
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	None defined				
Major 101	Load test of bridge rivet		0.40%	3.2	4.5.7.9
Minor 201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.37	Stud, Spring		8883721
			NEXT HIGHER ASSEMBLY 8883719

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	None defined				
Major 101	Diameter of dtud, min		0.40%	3.2	Gage
Minor 201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.5.7.8: Add "and removed from the lot".

PAGE 64

* 4.5.8, Delete in its entirety and substitute the following:

"4.5.8 Fuze body assembly seal leakage. The fuze body assembly shall be checked for leakage past the seal ring into the body assembly. The fuze body assembly shall be assembled to a test fixture, sealed, and submerged completely in water. A 5.0 psig plus or minus 0.25 psig shall be applied through the test fixture into the fuze body assembly. The integrity of the seal shall be verified by observation after 10 seconds up to a minimum of 30 seconds for the presence of a stream or recurring seccession of small bubbles. Any steady stream or recurring bubbles shall be classified as leakers and rejects. The test equipment calibration and equipment sensitivety shall be in accordance with paragraph 4.4.4."

PAGE 67

* 4.5.20, Delete in its entirety and substitute the following:

"4.5.20 Seal leakage test. The fuze shall be checked for leakage:

a. Load fuze onto a movable platform in an air tight bell jar.

b. Draw a vaccum slowly (1 to 2 inches of mercury/minute) until the vaccum drawn is equal to or slightly greater than the water pressure head on the fuze at its deepest depth.

c. Quickly submerge fuze by lowering platform in the water.

d. Further reduce vaccum to 10 inches of mercury and maintain constant by regulator for a minimum of 1 minute.

e. The integrity of the seal shall be verified by observation after 10 seconds up to a minimum of 30 seconds for the presence of a steady stream or recurring succession of bubbles. Any steady stream or recurring bubbles shall be classified as leakers/rejects.

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f. If big bubbles are seen, immediately life fuze from the water to prevent entry into the fuze.

g. After the observation period, life the fuze above the water and vent the vaccum through the regulator valve thus bleeding the bubble tester back to atmospheric pressure.

h. A leaker shall reject the fuze. The test equipment calibration and equipment sensitivity shall be in accordance with 4.4.4."

PAGE 70

* TABLE II: Delete "Firing Test" in its entirety.

Delete "Arming time over maximum 2 3"

Delete "Fail to arm 1 2"

Add: "Note: Following the drop test, the fuzes must remain safe in the out-of-line position but need not be operable."

4.5.29: Delete in its entirety and substitute:

"4.5.29 Non-Environmental Test - A sample of two hundred and nineteen (219) fuzes shall be selected and subjected to and pass the non-arm test and acceptance criteria specified in 4.5.31. If the test is successful, seventy-three (73) of the 219 fuzes shall be randomly selected and subjected to and pass the arming test and acceptance criteria specified in 4.5.31. The remaining 146 fuzes shall be equally divided into two groups of 73 each. One group shall be subjected to the Low Temperature Test specified in 4.5.35 and the remaining group shall be subjected to the High Temperature Test specified in 4.5.36."

Add:

* "4.5.35 Low Temperature Test - A sample of 73 fuzes shall be selected as specified in 4.5.29 and subjected to low temperature environment in accordance with Low Temperature Procedure I at -65°F, Method 502 of MIL-STD-810 with the exception that the time of exposure to temperature shall be a minimum of 4 hours and temperature

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inside the chamber shall be maintained at ± 5°F. The fuzes shall then be subjected to and pass arming time test and acceptance criteria specified in 4.5.31 within 5 minutes of removal from Low Temperature.

NOTE:

Removing the fuze from the cold temperature environment prior to performing the centrifuge test may cause frost to form on the metal parts. This may have an effect on the test results due to attenuation of the acoustic signal of the arming mechanism, thus making it difficult to distinguish the arm signal from background noise. The contractor shall take appropriate action to alleviate this potential problem. One effective method has been to seal each fuze in a plastic bag prior to temperature conditioning. Bagged fuzes are then subjected to the centrifuge test.

* 4.5.36 High Temperature Test - A sample of 73 fuzes shall be selected as specified in 4.5.29 and subjected to high temperature environment in accordance with High Temperature Procedure I, Method 501 of MIL-STD-810 with the exception that the time of exposure to temperature shall be a minimum of 4 hours and the relative humidity shall be maintained at less than 20%. The fuzes shall then be subjected to and pass arming time test and acceptance criteria specified in 4.5.31 within 5 minutes of removal from High Temperature."

Note: There is no requirement for humidity control of the 160°F chamber if the room ambient air which is heated is less than 100°F with a relative humidity below 95 percent and no water is added within the chamber during the test cycle. If these conditions cannot be met, the test chamber must be controlled to insure that the relative humidity is maintained below 20 percent."

PAGE 71

4.5.32: Delete subparagraph a in its entirety and substitute "a. Times under .63 seconds for the M423 and 1.07 seconds for the M427 are classed as arming time under minimum."; Delete subparagraph b in its entirety and substitute "b. Times .93 seconds through 1.30 seconds for the M423 and 1.47 seconds through 1.52 seconds for the M427 are classed as arming time over maximum."

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4.5.32: Delete subparagraph c in its entirety and substitute "c. Times over 1.30 for the M423 and 1.52 for the M427 or fuzes which do not arm or fuzes which arm but do not detent are classified as failure to arm."; Delete subparagraph e in its entirety and substitute "e. All fuzes which are not classed as a failure shall be radiographically inspected prior to firing (vibrated fuzes) or their return to the lot (non-environmental fuzes including hot and cold fuzes)."

PAGE 74

6.3: Delete "DRDAR-QAR-I" and substitute "DRSMC-QAF-I(D)".

6.5: Delete "DRDAR-QAM-M" and substitute "DRSMC-QAF-T(D)".

Delete "DRDAR-LCF" and substitute "DRSMC-LCN-M(D)".

6.8: Delete in its entirety and substitute:

"6.8 When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.
4.5.34	Quality Inspection Test, Demonstration and Evaluation Report	DI-R-1724
4.5.34	Quality Inspection Defect Report	DI-R-1721

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"(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Vol II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)"

"6.9 Source control and specification control drawings. Certificates of conformance from the supplier certifying that the item complies with the drawing requirements shall be provided to the Government representative."

*The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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